PART A

(25 marks)

1. Evaluate the following limits (if exists).

(a) $\lim_{x \to 4} \frac{x^4 - 16}{x - 2}$ [3 marks]

(b) $\lim_{x \to 4} \frac{2 - \sqrt{x}}{4 - x}$. [4 marks]

(c) $\lim_{x \to -\infty} \frac{3}{4x^2}$ [2 marks]

2. **(a)** By using the definition $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$, if f is a function with $f'(4) = 2$, find $\lim_{x \to 4} \frac{f(x) - f(4)}{\sqrt{x} - 2}$. [5 marks]

(b) Differentiate each of the following function with respect to *x*.

i) $f(x) = (\ln x)^3$ [2 marks]

ii) $f(x) = 3e^{3x}$ [2 marks]

iii) $f(x) = 4\sin^3 x$ [2 marks]

3. Given $y = 3x + \frac{12}{x}$, $x > 0$. Find the coordinates of the stationary point. Hence determine the nature of the point.

[5 marks]

PART B

(75 marks)

1. (a) Find integer values of *m* and *n* for which $m - n\log_3 2 = 10\log_9 6$

[4 marks]

(b) The complex numbers z_1 and z_2 are given by $z_1 = p + 2i$ and $z_2 = 1 - 2i$ where p is an integer

i) Find $\frac{z_1}{z_2}$ in the form of *a* + *bi* where *a* and *b* are real numbers.

[3 marks]

ii) Hence, find the possible values of *p* when $\left|\frac{z_1}{z_2}\right| = 13$. [4 marks]

2. Solve

a) $-2|2x+3|+14 \geq -16$ [3 marks]

b) $\left|\frac{x-1}{3x+1}\right| \geq 1$ [7 marks]

3. Given the functions $f(x) = 2 - x^2$ and $g(x) = x + 2$,find $f \circ g$ and $g \circ f$. Hence, determine the value of x such that $(f \circ g)(x) = (g \circ f)(x)$

[6 marks]

4. A function f is defined as $f(x) = 3 + \sqrt{x-2}$

i) Show that the function $f^{-1}(x)$ exists and hence, find $f^{-1}(x)$ [5 marks]

ii) State the domain and range of $f^{-1}(x)$. [2 marks]

iii) On the same axes, sketch the graphs of *f (x)* and $f^{-1}(x)$. [5 marks]

State the relationship between the two graphs.

5. a) Find the horizontal asymptote of $f(x)=\dfrac{2x}{(x+1)(x-2)}$. [4 marks]

b) The function f is defined by $f(x)=\begin{cases}\dfrac{x-1}{x+2}, & 0\le x<2\\ ax^2-1, & x\ge 2\end{cases}$. Find the value of a if $\lim_{x\to 2} f(x)$ exist. Hence, determine whether f is continuous at x = 2

[7 marks]

6. (a) A curve has an equation $x^2+y^2-2y=4$

i. Find $\dfrac{dy}{dx}$ in the terms of x and y.

ii. Determine the gradient of the curve at point (1,3)

iii. Express $\dfrac{d^2y}{dx^2}$ in terms of y.

[11 marks]

(b) The curve defined by the parametric equations, $x=t^2-3$ and $y=t^3+t$

Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ [5 marks]

7. A closed rectangular box has a base with its length twice its width, and the total surface area of the box is 300 cm^2. If the width of the base of the box is $\boldsymbol{x}$ cm, and the volume of the box is $\boldsymbol{V}$ cm^3. Show that $V=100x-\dfrac{4}{3}x^3$. Find the height of the box when its volume is maximum. Hence, find the maximum volume of the box.

[9 *marks*]

END OF QUESTIONS PAPER

ANSWERS

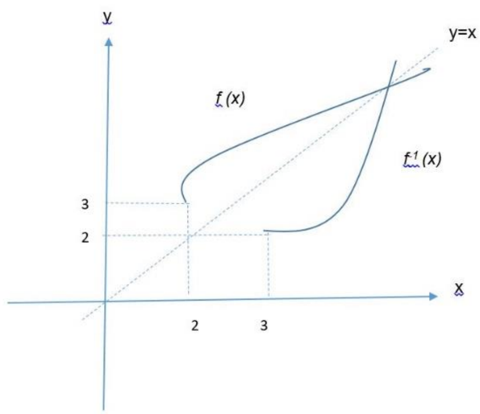
PART A

1. (a) 0 (b) $\frac{1}{4}$ (c) 0
2. (a) 8 (bi) $\frac{3}{x}(\ln x)^2$ (bii) $9e^{3x}$ (biii) $12\sin^2 x\cos x$
3. (2 , 12) , minimum point

PART B

1. (a) m = 5, n = -5 (bi) $\frac{p-4}{5}+\frac{2}{5}(p+1)i$ (bii) $p=\pm 29$
2. (a) $-9\le x\le 6$ (b) $\left[-1,-\frac{1}{3}\right)\cup\left(-\frac{1}{3},0\right]$
3. $x=-\frac{3}{2}$
4. (ai) $f^{-1}(x)=(x-3)^2+2$ (aii) $D_{f^{-1}}=[3,\infty)$, $R_{f^{-1}}=[2,\infty)$

   (aiii)



   $f^{-1}(x)$ is a reflection of graph f (x) about the line y = x.
5. (a) y = 0 (b) $a=\frac{5}{16}$, f (x) is continuous at x = 2
6. (ai) $\frac{x}{1-y}$ @ $\frac{-x}{y-1}$ (aii) $-\frac{1}{2}$ (aiii) $\frac{5}{(1-y)^2}$

   (b) $\frac{dy}{dx}=\frac{3t^2+1}{2t}$, $\frac{d^2y}{dx^2}=\frac{3t^2-1}{4t^3}$
7. $\frac{1000}{3}cm^3$